# Notes: Avramov, Ge, Li, and Linton (JFE, 2026)

**Dual Peer Effects and Cross-Stock Predictability**

## Contents

# 1 What the paper is doing

The paper builds an *economically structured* cross-stock return predictor called the *Peer Index* (PI). Instead of predicting a stock's return only from its own characteristics, PI uses a peer network (industry peers, shared-analyst peers, or customer–supplier peers) to map *peer information* into the stock's expected return. Conceptually, PI decomposes the information that matters for stock $i$ into two channels: (i) **peer strength** (how strong $i$'s peers are on average) and (ii) **within-peer position** (how unusual $i$ is relative to its peers). Empirically, PI predicts both stock returns and earnings surprises over short and long horizons. Importantly, even powerful machine-learning (ML) expected-return proxies that use *only own-stock characteristics* do not subsume PI, which supports the interpretation that PI captures *genuine cross-stock information* rather than merely a better nonlinear transformation of a stock's own signals. Finally, the paper provides dynamic evidence (lag-augmented local projections) consistent with *slow diffusion* of peer information into prices (positive PI shocks predict gradually decaying future returns without reversal).

# 2 Notation and objects

- Time $t$ is monthly (the main return-prediction setting).
- There are $N_t$ stocks at time $t$, indexed by $i$.
- $R_{i,t+1}$: (excess) return of stock $i$ from $t$ to $t+1$.
- $x_{i,t} \in \mathbb{R}^S$: a vector of $S$ firm-level characteristics for stock $i$ at time $t$. The paper uses a large set (Appendix Table 12).
- A peer network determines, for each stock $i$, a peer set $g_i$ (e.g., SIC-industry peers, shared-analyst peers, or customer–supplier peers).

# 3 Core framework: dual peer effects

## 3.1 Baseline return prediction

A general return-prediction problem can be written as:

$$R_{i,t+1} = h(x_{i,t}) + e_{i,t+1}, \tag{1}$$

where $h(\cdot)$ is the (possibly nonlinear) mapping from characteristics to expected returns.

A standard linear single-characteristic specialization is

$$R_{i,t+1} = x_{s,i,t}\,\beta + e_{i,t+1}, \qquad s \in \{1, \ldots, S\}. \tag{2}$$

### 3.2 Peer average and deviation-from-peers

Given a peer set $g_i$, define for each characteristic $s$:

$$z_{s,i,t} \equiv \frac{1}{|g_i|} \sum_{j \in g_i} x_{s,j,t} \qquad \text{(peer average / peer strength)}, \tag{3}$$

$$\tilde{x}_{s,i,t} \equiv x_{s,i,t} - z_{s,i,t} \qquad \text{(deviation from peers / within-peer position)}.$$

If one simply substitutes $x_{s,i,t} = z_{s,i,t} + \tilde{x}_{s,i,t}$ into Eq. (2), one obtains

$$R_{i,t+1} = z_{s,i,t}\,\beta + \tilde{x}_{s,i,t}\,\beta + e_{i,t+1}, \tag{4}$$

which mechanically imposes that the peer-average and deviation components enter with the *same* slope $\beta$.

### 3.3 Dual peer effects model

The key conceptual move of the paper is to *relax* that restriction and allow peer averages and deviations to have different pricing/predictive roles:

$$R_{i,t+1} = z_{i,t}'\beta_z + \tilde{x}_{i,t}'\beta_x + e_{i,t+1}, \tag{5}$$

where $z_{i,t} \in \mathbb{R}^S$ stacks $\{z_{s,i,t}\}_{s=1}^S$ and $\tilde{x}_{i,t} \in \mathbb{R}^S$ stacks $\{\tilde{x}_{s,i,t}\}_{s=1}^S$.

**Economic interpretation (the "two channels").**

- **Peer-average channel ($z_{i,t}$):** information embedded in the peer group. In an industry network, $z_{i,t}$ captures the *state of the industry* (e.g., peer profitability, investment, liquidity, etc.). This is naturally related to factor exposures implied by characteristics.
- **Deviation-from-peers channel ($\tilde{x}_{i,t}$):** information in how a firm differs from its peers. This captures *relative positioning* (e.g., "cheap vs expensive relative to industry", "high vs low investment relative to industry").

Both channels can forecast returns, and they need not do so with the same sign or magnitude.

## 4 Peer networks as a position matrix

### 4.1 Industry linkage matrix

For the baseline peer definition, the paper uses SIC-based industry peers. Let $L^{\text{ind}}$ be an $N \times N$ linkage matrix with entries

$$\left(L^{\text{ind}}\right)_{ij} \equiv \mathbb{I}\{i \text{ and } j \text{ in same industry}\} \cdot \frac{1}{|g_i|}. \tag{6}$$

This is a row-normalized adjacency matrix, so each row averages over $i$'s peers.

### 4.2 Matrix expression for peer averages

Let $x_{s,t} \in \mathbb{R}^N$ be the cross-section of characteristic $s$ at time $t$. Then for a general position matrix $L$,

$$z_{s,i,t} = x'_{s,t} L_i, \qquad \tilde{x}_{s,i,t} = x_{s,i,t} - z_{s,i,t}, \tag{7}$$

where $L_i$ is the $i$th column (or equivalently, $L_i$ extracts the appropriate weighted average for $i$).

# 5 How PI is constructed in practice

### 5.1 Ridge estimation of the dual peer effects mapping

Operationally, PI is the *fitted value* from a high-dimensional version of Eq. (5) using all $S$ characteristics:

$$R_{i,t+1} \approx z'_{i,t}\beta_z + \tilde{x}'_{i,t}\beta_x.$$

Because $S$ is large and the regressors are correlated, the paper uses ridge regularization.

A generic ridge objective for estimating $(\beta_z, \beta_x)$ is:

$$\min_{\beta_z,\beta_x} \sum_{(i,t)\in\mathcal{W}} \left(R_{i,t+1} - z'_{i,t}\beta_z - \tilde{x}'_{i,t}\beta_x\right)^2 + \lambda\left(\|\beta_z\|_2^2 + \|\beta_x\|_2^2\right), \tag{1}$$

where $\mathcal{W}$ denotes the estimation window and $\lambda > 0$ is a tuning parameter.

**PI signal.** Given estimated coefficients, define the raw PI forecast:

$$\widehat{R}^{PI}_{i,t+1} \equiv z'_{i,t}\widehat{\beta}_z + \tilde{x}'_{i,t}\widehat{\beta}_x.$$

The paper then standardizes / ranks PI each month so that the signal is comparable across time and interpretable in cross-sectional regressions and portfolio sorts.

### 5.2 Why PI is not "just ML"

A key comparison throughout the paper is between:

- ML predictors that only use *own-stock* characteristics $x_{i,t}$, and
- PI, which uses peer averages and peer deviations $(z_{i,t}, \tilde{x}_{i,t})$.

If PI remains predictive after controlling for ML expected-return proxies, that is interpreted as evidence that PI captures *cross-stock* information content that cannot be recovered from $x_{i,t}$ alone.

# 6 Connection to principal portfolios and attention-induced pricing

### 6.1 Attention-induced pricing model (AIPM)

The paper notes an AIPM-style portfolio-weight mapping:

$$w^{AIPM}_{i,t} = \lambda' X'_t A_i, \tag{8}$$

where $X_t$ stacks characteristics, $A$ is an "attention" (or position) matrix, and $A_i$ is the relevant column.

With $A$ as the position matrix, one can write

$$z_{s,i,t} = \sum_{j=1}^{N} A_{ij} x_{s,j,t} = A_i' x_{s,t}, \qquad z_{i,t} = X_t' A_i. \tag{9}$$

If $\lambda = \beta_z$, then

$$w_{i,t}^{AIPM} = \beta_z' z_{i,t}. \tag{10}$$

This highlights that the peer-average leg can be viewed as an economically motivated "position" in characteristic space.

### 6.2 Bayesian shrinkage toward industry linkages

Rather than fully learning $A$ from scratch, a natural disciplined approach is to shrink toward the industry linkage matrix:

$$A_{ij} \sim \mathcal{N}\left(\left(L^{\text{ind}}\right)_{ij}, \tau^2\right). \tag{11}$$

This makes the industry network a prior mean and allows data-driven deviations.

## 7 Empirical designs and evaluation metrics

### 7.1 Out-of-sample performance: $R^2$ and prediction-error comparisons

Out-of-sample performance for model $m$ is measured by

$$R^2_{OOS} = 1 - \frac{\sum_t \sum_i \left(R_{i,t+1} - \widehat{R}^m_{i,t+1|t}\right)^2}{\sum_t \sum_i R^2_{i,t+1}}, \tag{12}$$

with returns de-meaned appropriately (as in the paper).

To compare models by their mean squared prediction error (MSPE), the paper uses

$$d_t^{m,m'} \equiv \sum_i \left(R_{i,t+1} - \widehat{R}^m_{i,t+1|t}\right)^2 - \sum_i \left(R_{i,t+1} - \widehat{R}^{m'}_{i,t+1|t}\right)^2, \tag{13}$$

and tests whether the average $d_t^{m,m'}$ differs from zero (Diebold–Mariano logic).

### 7.2 Peer Index factor (PIF) used for factor spanning

To study whether PI can be spanned by standard factors, the paper forms a *peer index factor* (PIF) by averaging PI long–short spreads in big and small stocks:

$$PIF_t \equiv \frac{1}{2} PIF_t^{big} + \frac{1}{2} PIF_t^{small}. \tag{14}$$

This construction is designed to mitigate concerns that predictability is mechanically driven by microcaps while retaining broad coverage.

### 7.3 Dynamics: lag-augmented local projections (LA-LP)

To study the dynamic return response to a PI shock, the paper estimates:

$$R_{i,t+h} = \alpha^{(h)} + \beta^{(h)} PI_{i,t} + \sum_{j=1}^{12} \phi_j^{(h)} R_{i,t-j} + \sum_{j=1}^{12} \psi_j^{(h)} PI_{i,t-j} + \nu_{i,t}^{(h)}, \tag{15}$$

and plots $\beta^{(h)}$ across horizons $h$.

**Interpretation.** If PI captures *underreaction* to peer information, then a positive PI innovation should predict *positive future returns* that decay *gradually* and should not reverse strongly.

### 7.4 Fundamentals (earnings surprises) regressions

To connect PI to fundamentals, the paper studies whether PI predicts standardized unexpected earnings (SUE). At the industry level:

$$SUE_{ind,i,t,t+1} = \alpha + \beta\, PI_{i,t} + \gamma\, ML_{i,t} + \Gamma' W_{i,t} + \varepsilon_{i,t+1}, \tag{16}$$

and similarly for industry-level returns.

At the firm level, for horizon $h$:

$$SUE_{i,t+h} = \alpha + \beta\, PI_{i,t} + \gamma\, ML_{i,t} + \Gamma' W_{i,t} + \varepsilon_{i,t+h}. \tag{17}$$

## 8 Main results: interpreting each table and figure

### 8.1 Table 1: Fama–MacBeth cross-sectional regressions

**Object:** monthly value-weighted Fama–MacBeth regressions of $R_{i,t+1}$ on PI and standard predictors.

- **Panel A (basic controls):** PI enters with a large, positive, and statistically significant slope after controlling for short-term reversal and momentum, and after controlling for either industry momentum or a lead–lag term. Interpretation: the signal is not a relabeling of standard momentum/reversal or simple industry momentum.
- **Panel B (robustness):** PI remains significant when controlling for ML expected-return proxies. Interpretation: own-characteristic ML does not subsume the peer-based signal; this supports the "genuine cross-stock information" view.
- **Panel C (window settings):** PI remains predictive under alternative estimation-window choices. Interpretation: the result is not fragile to a particular training-window design.

- **Panel D (long-horizon predictability):** PI predicts multi-month cumulative returns. Interpretation: predictability is not purely one-month reversal; it persists.
- **Panel E (market state):** PI predicts in both "high" and "low" states. Interpretation: PI is not only a crisis artifact.
- **Panel F (nonlinearity):** a nonlinear version of PI has additional explanatory power. Interpretation: the PI mapping can benefit from mild nonlinearity, but the core peer logic remains.

**Table 1**
Value-weighted Fama–MacBeth cross-sectional regressions.

Panel A: Basic controls

| Dependent variable | Observations | PI | MOM1 | MOM12 | INDMOM | Lead-lag |
|---|---|---|---|---|---|---|
| $R_{t+1}$ | 2,088,332 | | −0.68***<br>(−3.19) | 0.83**<br>(2.19) | 0.51***<br>(3.15) | |
| $R_{t+1}$ | 2,088,332 | 0.99***<br>(5.90) | −0.24<br>(−1.15) | 0.54<br>(1.44) | 0.06<br>(0.36) | |
| $R_{t+1}$ | 2,088,332 | | −0.78***<br>(−3.78) | 0.86**<br>(2.26) | | 0.60***<br>(3.56) |
| $R_{t+1}$ | 2,088,332 | 0.85***<br>(5.71) | −0.41**<br>(−2.00) | 0.60<br>(1.60) | | 0.33*<br>(1.80) |

Panel B: Robustness

| Dependent variable | | Observations | PI |
|---|---|---|---|
| $R_{t+1}$ | With ML control (GBRT) | 2,088,332 | 0.81***<br>(4.29) |
| $R_{t+1}$ | With ML control (RF) | 2,088,332 | 0.78***<br>(4.17) |
| $R_{t+1}$ | With ML control (NN) | 2,088,332 | 0.61***<br>(3.15) |
| $R_{t+1}$ | With all controls | 2,088,332 | 0.50***<br>(3.87) |
| $R_{t+1}$ | Remove microcaps | 1,043,044 | 0.92***<br>(4.61) |
| $R_{t+1}$ | Post 2003 Sample | 942,090 | 1.96***<br>(7.47) |

Panel C: Window settings

| Dependent variable | Window size | Observations | PI | PI post 2003 |
|---|---|---|---|---|
| $R_{t+1}$ | Rolling 12 | 2,088,332 (942,090) | 2.57***<br>(8.66) | 1.46***<br>(5.85) |
| $R_{t+1}$ | Rolling 60 | 2,088,332 (942,090) | 2.81***<br>(9.64) | 1.35***<br>(5.72) |
| $R_{t+1}$ | Rolling 120 | 2,088,332 (942,090) | 3.00***<br>(10.51) | 1.57***<br>(6.69) |
| $R_{t+1}$ | Rolling 360 | 2,088,332 (942,090) | 3.26***<br>(11.60) | 1.96***<br>(7.31) |
| $R_{t+1}$ | Expanding | 2,088,332 (942,090) | 3.28***<br>(11.52) | 1.96***<br>(7.47) |

Panel D: Long-term predictability

| Dependent variable | Observations | PI |
|---|---|---|
| $R_{t+1:t+6}$ | 2,088,332 | 2.43***<br>(2.62) |
| $R_{t+1:t+12}$ | 2,088,332 | 3.61**<br>(2.19) |
| $R_{t+1:t+36}$ | 2,088,332 | 8.16***<br>(2.57) |

Panel E: Market state

| Dependent variable | | Observations high state | PI high state | Observations low state | PI low state |
|---|---|---|---|---|---|
| $R_{t+1}$ | Volatility | 863,531 | 1.07***<br>(2.80) | 779,723 | 0.93***<br>(3.94) |
| $R_{t+1}$ | Liquidity | 1,027,966 | 1.04***<br>(4.07) | 1,097,253 | 0.96***<br>(2.95) |
| $R_{t+1}$ | Sentiment | 1,046,014 | 1.39***<br>(5.35) | 969,629 | 0.75**<br>(2.48) |

Panel F: Nonlinear effect

| Dependent variable | | Observations | PInonlinear | PI |
|---|---|---|---|---|
| $R_{t+1}$ | Feature-first transformation | 2,088,332 | 0.75***<br>(4.51) | |
| $R_{t+1}$ | Feature-first transformation | 2,088,332 | 0.49***<br>(2.90) | 0.88***<br>(4.40) |
| $R_{t+1}$ | Peer-first transformation | 2,088,332 | 1.08***<br>(5.99) | |
| $R_{t+1}$ | Peer-first transformation | 2,088,332 | 0.64***<br>(2.93) | 0.59**<br>(2.40) |

Panels A–F report average slopes and Newey–West $t$-statistics (12 lags) from monthly value-weighted Fama–MacBeth regressions. Monthly returns are expressed in percent. MOM1 and MOM12 follow Jegadeesh (1990) and Jegadeesh and Titman (1993). INDMOM is the industry momentum variable from Moskowitz and Grinblatt (1999). Lead–Lag is the industry lead–lag signal from Hong et al. (2007). Panel E examines the PI's predictability across market states; "low" ("high") states are months below (above) the cross-sectional median of the previous month's value of the state variable (VIX for volatility, Amihud (2002) for illiquidity, and Baker and Wurgler (2006) for sentiment). Panel F evaluates a nonlinear version of PI constructed from quadratic and cubic transformations of peer-average and peer-deviation characteristics. The sample period is January 1980–December 2024. In the main sample, we exclude "nano" stocks (market equity below the NYSE 1st percentile). The "no microcaps" sample excludes firms below the NYSE 20th percentile of market equity. One, two, and three asterisks denote significance at the 10%, 5%, and 1% levels, respectively.

**Table 2**
Monthly out-of-sample performance.

| | NN | GBRT | RF |
|---|---|---|---|
| $R2\_OOS^{ML}$ | 0.322% | 0.242% | 0.306% |
| $R2\_OOS^{PI}$ | 0.325% | 0.325% | 0.325% |
| $R2\_OOS^{AVE}$ | 0.498% | 0.391% | 0.427% |
| DM | 2.387 | 2.153 | 1.740 |
| p-value | (0.017) | (0.031) | (0.082) |

This table reports out-of-sample performance for three machine-learning methods: neural network (NN), gradient-boosted regression trees (GBRT), and random forest (RF). For each method, we report $R^2_{\text{OOS}}$ for: (1) a benchmark using 130 firm-specific characteristics ($ML$); (2) a model using only the Peer Index ($PI$); and (3) a simple average of the $ML$ and $PI$ forecasts ($AVE$). The ML models use an expanding-window scheme with an initial 60-month training period (1973–1977) and a 24-month validation period (1978–1979), with the out-of-sample period starting in January 1980. The $PI$ model is estimated with a rolling 84-month window, yielding an out-of-sample start of January 1987. "DM" reports the modified Diebold–Mariano statistic (Gu et al., 2020 adjustment) comparing $AVE$ vs. $ML$; Newey–West standard errors are used and $p$-values appear in parentheses.

## 8.2 Table 2: Out-of-sample performance vs ML

**Object:** compare out-of-sample $R^2$ across (i) ML using only $x_{i,t}$, (ii) PI using $(z_{i,t}, \tilde{x}_{i,t})$, and (iii) the simple average forecast.

- PI alone has slightly higher $R^2_{OOS}$ than each ML benchmark in the table.
- The average forecast (ML+PI) improves $R^2_{OOS}$ further, with statistically significant MSPE improvements (Diebold–Mariano tests).

**Interpretation:** PI contains predictive information that is complementary to own-characteristic ML forecasts.

## 8.3 Table 3: Portfolio sorts on PI (equal- and value-weighted)

**Object:** sort stocks into PI quintiles and form top-minus-bottom portfolios.

- Equal-weighted spreads are very large and remain highly significant after controlling for standard factor models.
- Value-weighted spreads are smaller (consistent with stronger effects in smaller firms) but remain positive and statistically significant.
- "ML-adjusted" PI sorts still deliver significant spreads.

**Table 3**
Risks and alphas of PI-based portfolios.

| | Excess ret | Alpha | Market | SMB | HML | RMW | CMA | UMD | REV |
|---|---|---|---|---|---|---|---|---|---|
| Panel (A) Equal-weighted PI portfolios (7-factor model) | | | | | | | | | |
| Top-minus-bottom | 3.33*** | 3.18*** | 0.03 | 0.10** | 0.06 | −0.14** | 0.38*** | 0.25*** | −0.18*** |
| | (23.03) | (23.60) | (1.01) | (2.15) | (1.01) | (−2.25) | (4.30) | (7.75) | (−4.34) |
| Top-minus-bottom (post 2003) | 2.25*** | 2.24*** | −0.05 | 0.02 | 0.09 | 0.16* | 0.39*** | 0.07* | −0.06 |
| | (13.44) | (13.81) | (−1.17) | (0.34) | (1.21) | (1.80) | (3.60) | (1.70) | (−0.98) |
| Top-minus-bottom (adjust ML) | 1.87*** | 1.79*** | 0.10*** | −0.03 | 0.17*** | −0.11* | 0.25*** | 0.18*** | −0.41*** |
| | (12.94) | (14.00) | (3.26) | (−0.63) | (3.01) | (−1.88) | (2.98) | (6.03) | (−10.43) |
| Panel (B) Value-weighted PI portfolios (7-factor model) | | | | | | | | | |
| Top-minus-bottom | 0.89*** | 0.68*** | 0.07** | −0.08* | 0.21*** | −0.20*** | 0.20** | 0.41*** | −0.18*** |
| | (6.05) | (5.45) | (2.47) | (−1.78) | (3.88) | (−3.59) | (2.43) | (14.25) | (−4.58) |
| Top-minus-bottom (post 2003) | 0.34* | 0.34** | −0.02 | −0.17** | 0.30*** | 0.04 | 0.30*** | 0.36*** | −0.01 |
| | (1.73) | (2.12) | (−0.40) | (−2.39) | (4.20) | (0.41) | (2.81) | (8.78) | (−0.18) |
| Top-minus-bottom (adjust ML) | 0.53*** | 0.40*** | 0.09*** | −0.16*** | 0.22*** | −0.11* | 0.14* | 0.24*** | −0.26*** |
| | (3.76) | (3.11) | (3.00) | (−3.53) | (3.86) | (−1.85) | (1.62) | (8.12) | (−6.63) |

This table reports the monthly excess returns, factor loadings, and alphas of PI-based portfolios, considering a seven-factor model includes the Fama–French five factors plus the momentum factor and the reversal factor. Panel A and B show the results for the equal-weighted and value-weighted portfolios, respectively. The sample period is Jan 1980-Dec 2024. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

**Table 4**
Annual alphas of PI-portfolios over longer holding periods.

| | Holding period | | |
|---|---|---|---|
| | 3 | 6 | 12 |
| VW top quintile | 0.36 | 0.85 | 0.79* |
| | (0.49) | (1.49) | (1.73) |
| VW bottom quintile | −4.01*** | −2.47*** | −1.44*** |
| | (−4.84) | (−3.68) | (−2.82) |
| VW spread portfolios | 4.37*** | 3.31*** | 2.23*** |
| | (7.46) | (8.77) | (8.67) |
| VW spread portfolios (post 2003) | 2.12*** | 1.63*** | 1.28*** |
| | (2.84) | (3.15) | (3.84) |

This table reports the annual alphas (in percentage) and their t-values (in parentheses) obtained from regressing monthly value-weighted PI portfolio returns on Fama–French five factors plus the momentum factor and the reversal factor. Annual alphas are obtained by multiplying monthly alphas by 12. We follow the technique from Jegadeesh and Titman (1993) to deal with holding periods longer than one month. The sample period is Jan 1980 to Dec 2024. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

**Interpretation:** the PI signal is economically meaningful in trading-portfolio space and not simply an artifact of known factor exposures.

## 8.4 Figure 1: Cumulative performance of PI portfolios

**Object:** cumulative returns of the PI strategy relative to the market.
**Interpretation:** the PI-based long–short portfolio delivers persistent cumulative gains, indicating that the predictive content is stable enough to accumulate economically meaningful performance.

## 8.5 Table 4: Longer holding periods

**Object:** annualized alphas for holding PI portfolios over 3-, 6-, and 12-month horizons.
**Interpretation:** the spread alphas remain strongly positive even for longer holding periods, which is consistent with *slow diffusion* (not quick reversal).

## 8.6 Figure 2: Model complexity vs realized OOS performance

**Object:** how portfolio return/Sharpe and model $R^2_{OOS}$ vary with model complexity $c = P/T$.
**Interpretation:** performance is not monotone in complexity; disciplined regularization matters. The figure supports that the PI construction balances fit and stability in a way that performs well out of sample.

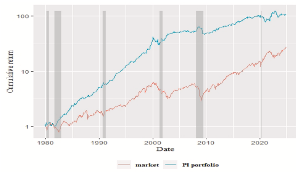



**Fig. 1.** PI-based portfolio. This figure plots the value of portfolios over time after investing $1 at the beginning of the sample period. The PI portfolio represer the value-weighted top-minus bottom PI-based portfolio. The market is the CRSP value-weighted index. The sample period is Jan 1980 to Dec 2024. Shaded are show the periods of recessions from Federal Reserve Economic Data (FRED). (For interpretation of the references to color in this figure legend, the reader referred to the web version of this article.)

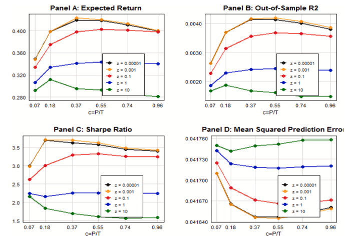



**Fig. 2.** Realized out-of-sample PI-based portfolio average return, Sharpe ratio, PI-based model out-of-sample $R^2$, and mean squared prediction error. The horizontal axis shows model complexity $c = P/T$, with $P \in \{20, 50, 100, 200, 260\}$ and $T$ is the average window size across all expanding windows.

**Table 5**
Alpha and factor loadings.

| Alpha | Market | SMB | HML | RMW | CMA | UMD | REV | LIQ | PEAD | FIN | R_IA | R_ROE | MGMT | PERF |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Panel (A): PIF | | | | | | | | | | | | | | |
| 1.00*** | −0.18*** | −0.28** | 0.13 | 0.37*** | 0.25 | | | | | | | | | |
| (3.32) | (−2.59) | (−2.40) | (1.18) | (2.61) | (1.35) | | | | | | | | | |
| 0.74*** | −0.02 | −0.14 | 0.24*** | 0.45*** | 0.15 | 0.31*** | −0.10 | −0.02 | | | | | | |
| (2.92) | (−0.33) | (−1.42) | (2.60) | (3.77) | (0.96) | (6.31) | (−1.07) | (−0.62) | | | | | | |
| 1.06*** | −0.19* | | | | | | | | 0.45*** | 0.03 | | | | |
| (3.19) | (−1.95) | | | | | | | | (2.72) | (0.19) | | | | |
| 0.82*** | −0.03 | −0.07 | | | | | | | | | 0.49*** | 0.54*** | | |
| (3.23) | (−0.51) | (−0.64) | | | | | | | | | (4.36) | (5.45) | | |
| 0.94*** | 0.02 | −0.04 | | | | | | | | | | | 0.31** | 0.37*** |
| (2.93) | (0.23) | (−0.33) | | | | | | | | | | | (1.93) | (4.69) |
| 0.87** | −0.09 | −0.16 | 0.28 | 0.78*** | −0.64 | 0.35*** | 0.08 | 0.02 | −0.07 | −0.38** | 0.71* | −0.26 | 0.23 | 0.09 |
| (2.46) | (−0.62) | (−1.26) | (1.38) | (3.50) | (−1.59) | (3.14) | (0.44) | (0.53) | (−0.35) | (−2.33) | (1.70) | (−1.12) | (0.97) | (0.52) |
| Panel (B): PIF adjusted for machine learning | | | | | | | | | | | | | | |
| 0.95*** | 0.12*** | −0.09** | 0.12* | −0.16* | −0.10 | 0.12*** | −0.34*** | 0.02 | −0.04 | −0.12* | 0.27* | −0.03 | 0.22** | 0.12** |
| (7.29) | (3.82) | (−1.97) | (1.74) | (1.70) | (−0.63) | (2.88) | (−8.89) | (0.85) | (−0.57) | (−1.74) | (1.79) | (−0.37) | (2.55) | (1.98) |

This table reports the alphas, factor loadings, and their t-values (in parentheses) obtained from the factor-spanning tests of regressing the PI factor (PIF) on other factors. Panel (A) presents the results for the unadjusted PIF, while Panel (B) shows the results for the machine learning (ML)-adjusted PIF, where the predictive component driven by firm-specific characteristics is removed. The factors include FF5 factor, momentum factor (UMD), short-term reversal factor (REV), liquidity factor (LIQ) from Pástor and Stambaugh (2003), short-horizon inattention factor (PEAD) and long-horizon financing factor (FIN) from Daniel et al. (2020), investment factor (R_IA) and return on equity factor (R_ROE) from Hou et al. (2015), management factor (MGMT) and performance factor (PERF) from Stambaugh and Yuan (2017). PEAD and FIN are available from January 1980 to December 2018. MGMT and PERF are available from January 1980 to December 2016. All other factors are available during the whole sample period. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

## 8.7 Table 5: Factor spanning tests for PIF

**Object:** regress the PI factor (PIF) on prominent factor sets (FF5; FF5+UMD+REV+LIQ; Daniel et al. behavioral factors; HXZ; Stambaugh–Yuan; and a comprehensive factor set).

- PIF has a positive and significant alpha across many factor sets.
- Even in a rich "all factors" regression, PIF retains a statistically significant alpha.
- The ML-adjusted PIF (removing the own-characteristic ML component) also retains a large alpha.

**Interpretation:** PI is not fully spanned by common factor models; it looks like an incremental pricing/reaction phenomenon tied to peer information.

## 8.8 Figure 3 and Eq. (15): dynamic response (LA-LP)

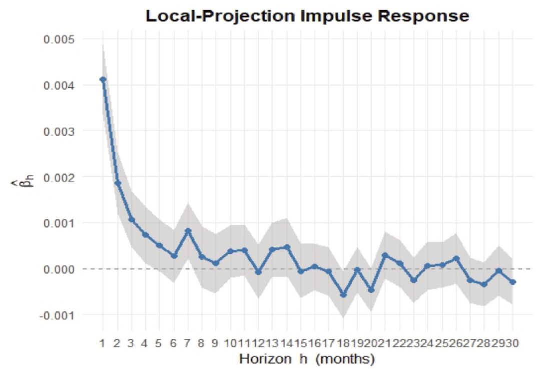
**Object:** impulse-response style evidence: $\beta^{(h)}$ from Eq. (15) across horizons $h$.
**Interpretation:** a positive PI innovation predicts positive future returns that decay gradually without reversal. This is the signature the authors emphasize for *slow diffusion of peer information* rather than overreaction.

## 8.9 Table 6: PI predicts industry-level earnings surprises and returns

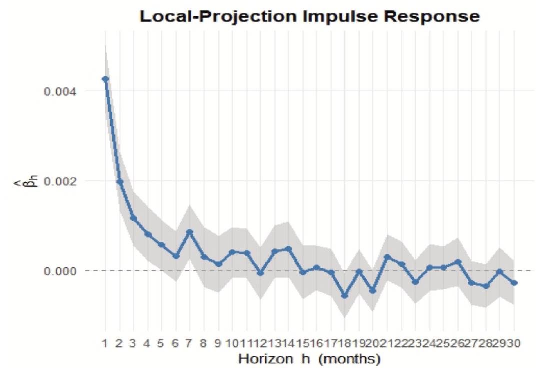
**Object:** regress industry-level SUE and industry returns on PI and ML.
**Interpretation:** PI strongly predicts industry-level earnings surprises and industry-level returns, while ML (own-characteristic) adds less for industry returns. This supports a mechanism where PI captures peer/industry information that is gradually incorporated into prices.

## 8.10 Table 7: PI predicts firm-level earnings surprises over multiple horizons

**Object:** firm-level SUE at horizons $h = 1, 4, 8$ (quarters) on PI and ML.
**Interpretation:** PI predicts near- and longer-horizon earnings surprises. In contrast, ML is strong at the short horizon but weak or negative at longer horizons. This pattern is consistent with PI capturing slower-moving peer-information diffusion into fundamentals.




(a) Panel A: Impulse response of $PI_{i,t}$ on future returns $R_{i,t+h}$ with a 95% confidence interval for horizons $h = 1$ to 30 months. PI is scaled by 10.




(b) Panel B: Impulse response of $PI_{i,t}$ shocks on future returns $R_{i,t+h}$ with a 95% confidence interval for horizons $h = 1$ to 30 months. The shock variable is scaled by 10.

**Fig. 3.** Lag-Augmented Local Projection (LA-LP).

**Table 6**
Predicting industry-level fundamentals & Returns.

| | $SUE_{ind,i,t+1}$ | $R_{ind,i,t+1}$ |
|---|---|---|
| PI | 0.004*** | 0.920*** |
| | (22.27) | (8.61) |
| ML | 0.001*** | 0.165 |
| | (5.77) | (1.52) |
| Controls | Yes | Yes |
| Fixed effect | Yes | Yes |
| Adjusted $R^2$ | 0.067 | 0.112 |
| Observations | 639,525 | 2,086,995 |

This table reports the results of panel regression for predicting one-step-ahead earnings surprises and excess returns at the industry level. The earnings surprise is measured by the standardized unexpected earnings (SUE), which is computed as the I/B/E/S actual earnings minus median forecasted earnings divided by fiscal-quarter-end market cap. If earning forecast is not available, then the unexpected earning is computed as the I/B/E/S actual earnings minus the seasonally differenced quarterly earnings. In the regression for predicting one-step-ahead earnings surprises, we include both time and entity fixed effects. Control variables include current industry average SUE, together with industry average size and book-to-market. $t$-statistics based on standard errors clustered by entity are shown below coefficient estimates. The second column of the table shows the average slopes*100; $t$-statistics with Newey–West adjustments of 12 lags from monthly Fama–MacBeth regressions for industry return. Control variables include lagged industry-average returns, industry-average size, and the book-to-market ratio. All explanatory variables are cross-sectionally standardized. The adjusted $R^2$ is the average across all cross-sectional regressions. For brevity, the coefficients on these controls are not reported. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

## 8.11 Table 8: where PI is strongest (information environment heterogeneity)

**Object:** interaction regressions showing whether PI is stronger for: smaller firms, fewer analysts, higher forecast dispersion, less news coverage, and smaller changes in news coverage.

**Interpretation:** PI is stronger exactly when attention and information processing are plausibly weaker, supporting an underreaction channel.

**Table 7**
Predicting firm-level earnings surprises.

| | (1) One-quarter-ahead | (2) Four-quarter-ahead | (3) Eight-quarter-ahead |
|---|---|---|---|
| PI | 0.026*** | 0.0112*** | 0.001*** |
| | (21.29) | (9.28) | (2.90) |
| ML | 0.027*** | −0.009*** | −0.008*** |
| | (16.74) | (−8.00) | (−6.55) |
| Observations | 639,525 | 608,144 | 556,180 |
| $Adj.R^2$ | 0.032 | 0.028 | 0.023 |
| Controls | Yes | Yes | Yes |
| Firm fixed effect | Yes | Yes | Yes |
| Time fixed effect | Yes | Yes | Yes |

This table reports the results of panel regression for predicting multiple-step-ahead earnings surprises. The earnings surprise is measured by the standardized unexpected earnings (SUE), which is computed as the I/B/E/S actual earnings minus the median forecasted earnings divided by fiscal-quarter-end market cap. If earning forecast is not available, then the unexpected earning is computed as the I/B/E/S actual earnings minus the seasonally differenced quarterly earnings. Control variables include the firm's current $SUE_{i,t}$ and its industry-average counterpart, along with size and the book-to-market ratio. Both regressions incorporate individual and quarter-fixed effects. All explanatory variables are cross-sectionally standardized. For brevity, the coefficients on these controls are not reported. $t$-statistics based on standard errors clustered by entity are shown below coefficient estimates. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

**Table 8**
Cross-sectional variation in PI effect.

| Dependent variable: One-month-ahead return | | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| PI | 0.93*** | 2.46*** | 2.87*** | 2.39*** | 1.52*** | 1.64*** |
| | (3.68) | (10.04) | (8.11) | (7.97) | (5.44) | (5.73) |
| PI*Dummy(Size < 30%) | 2.27*** | 2.13*** | | | | |
| | (4.10) | (10.95) | | | | |
| PI*Dummy(Analyst < 30%) | 0.10 | | 1.20*** | | | |
| | (0.22) | | (5.73) | | | |
| PI*Dummy(Dispersion > 70%) | 0.28 | | | 0.82*** | | |
| | (1.10) | | | (4.98) | | |
| PI*Dummy(News < 30%) | 0.92*** | | | | 1.28*** | |
| | (3.16) | | | | (5.97) | |
| PI*Dummy($\Delta$News < 30%) | 0.42* | | | | | 0.83*** |
| | (1.69) | | | | | (3.84) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Average# of stocks | 3895 | 3963 | 3963 | 3963 | 3895 | 3895 |
| Average adjusted $R^2$ | 0.051 | 0.036 | 0.034 | 0.042 | 0.040 | 0.040 |

This table reports regression results of the heterogeneity tests. We use five variables to proxy the speed of information flow, including size, number of analysts, analyst dispersion, news mentions and change in news mentions. The two news-based attention variables are available since 2000. For size, number of analysts, news, and $\Delta$ news, we create an indicator variable equal to 1 if the stock is below the bottom 30%, and 0 otherwise. For analyst dispersion, we create an indicator variable equal to 1 if the stock is above the top 30%, and 0 otherwise. In columns 2–6, we conduct FM regression by regressing future returns on PI and one of the interaction terms. In column 1, we conduct FM regression by regressing future returns on PI and all five interaction terms. Control variables include past one-month return, past 12-month return (excluding the most recent month), size and book-to-market. FM coefficients are the means of the monthly estimated coefficients*100; $t$-statistics employ Newey–West adjustments of 12 lags.

**Table 9**
Cross firm momentum effects and Fama–MacBeth cross-sectional regressions.

| Dependent variable | Observations | PI | INDMOM | Lead-lag | $SA\_Ret$ | $Cus\_Ret$ |
|---|---|---|---|---|---|---|
| $R_{t+1}$ | 2,088,332 | | 0.21 | | | |
| | | | (1.40) | | | |
| $R_{t+1}$ | 2,088,332 | 0.98*** | 0.07 | | | |
| | | (6.53) | (0.39) | | | |
| $R_{t+1}$ | 2,088,332 | | | 0.56** | | |
| | | | | (3.52) | | |
| $R_{t+1}$ | 2,088,332 | 0.86*** | | 0.31* | | |
| | | (6.45) | | (1.72) | | |
| $R_{t+1}$ | 1,674,481 | | | | 1.03*** | |
| | | | | | (4.62) | |
| $R_{t+1}$ | 1,674,481 | 0.84*** | | | 0.86*** | |
| | | (7.20) | | | (3.81) | |
| $R_{t+1}$ | 2,000,816 | | | | | 0.39*** |
| | | | | | | (2.90) |
| $R_{t+1}$ | 2,000,816 | 0.72*** | | | | 0.36*** |
| | | (5.50) | | | | (2.60) |
| $R_{t+1}$ | 1,674,481 | 0.87*** | −0.57* | 0.13 | 0.90*** | 0.24 |
| | | (5.92) | (−1.88) | (0.46) | (3.89) | (1.61) |

The table shows the average slopes*100; $t$-statistics with Newey–West adjustments of 12 lags from monthly value-weighted Fama–MacBeth regressions. Control variables include size, book-to-market ratio, short-term reversal, and momentum. The sample period is January 1980-Dec 2024. For the main sample, we exclude "nano" stocks whose market capitalization is below the 1st percentile of the NYSE sample. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

### 8.12 Table 9: relation to cross-firm momentum spillovers

**Object:** compare PI to known cross-firm signals: industry momentum, lead–lag via shared analysts, and customer momentum.
**Interpretation:** PI remains robust when controlling for these signals and partly subsumes some of them. Thus PI is not merely a repackaging of previously proposed cross-stock momentum spillovers.

# 9 Asymmetric networks and the symmetric–antisymmetric decomposition

## 9.1 Decomposing a directed network

For an asymmetric (directed) network with position matrix $L$, define

$$L_{\text{sym}} = \frac{1}{2}(L + L'), \qquad L_{\text{asym}} = \frac{1}{2}(L - L'), \qquad L = L_{\text{sym}} + L_{\text{asym}}. \tag{18}$$

Applying $L$ to a signal vector $x_t \in \mathbb{R}^N$ yields two peer-average legs:

$$z_{\text{sym}} = L_{\text{sym}} x_t, \qquad z_{\text{asym}} = L_{\text{asym}} x_t. \tag{19}$$

The first corresponds to symmetric peer exposure (a beta-style leg); the second captures cross-directional timing/alpha embedded in the directed part of the network.

For peer-adjusted signals, let $M = I - L$. Then

$$M_{\text{sym}} = \frac{1}{2}(M + M') = I - L_{\text{sym}}, \qquad M_{\text{asym}} = \frac{1}{2}(M - M') = -L_{\text{asym}}. \tag{20}$$

Hence the peer-adjusted strategy decomposes as

$$(I - L)x_t = (I - L_{\text{sym}})x_t + (-L_{\text{asym}})x_t. \tag{21}$$

## 9.2 Table 10: economic meaning of the four components

Table 10 summarizes the interpretation:

- $L_{\text{sym}} x_t$: symmetric peer average $\Rightarrow$ characteristic-implied factor beta.
- $L_{\text{asym}} x_t$: antisymmetric peer average $\Rightarrow$ factor-neutral, cross-directional timing/alpha.
- $(I - L_{\text{sym}})x_t$: symmetric peer deviation $\Rightarrow$ beta to peer-adjusted characteristics.
- $(I - L_{\text{asym}})x_t$: mixture $\Rightarrow$ own-signal beta plus alpha from peer-adjusted strategy.

## 9.3 Table 11: drop-one-out importance

Table 11 drops one component at a time and re-computes OOS $R^2$ and portfolio returns.

**Table 10**
PI components under an asymmetric network.

| Component | Type | Economic meaning |
|---|---|---|
| $L_{\text{sym}}\boldsymbol{x}_t$ | Symmetric | Beta on factor implied by the characteristic |
| $L_{\text{asym}}\boldsymbol{x}_t$ | Antisymmetric | Alpha/cross-directional timing |
| $(I - L_{\text{sym}})\boldsymbol{x}_t$ | Symmetric | Beta on factor implied by the peer-adjusted characteristic |
| $(I - L_{\text{asym}})\boldsymbol{x}_t$ | Mixture | Own-signal beta + alpha from peer-adjusted strategy |

**Table 11**
Drop-one-out analysis.

| Specification | (A) Analyst-network | | (B) Customer–supplier | |
|---|---|---|---|---|
| | OOS $R^2$ (%) | Port_ret (%) | OOS $R^2$ (%) | Port_ret (%) |
| Drop_sym_peer | 0.2498 | 2.898 | 0.1798 | 2.467 |
| Drop_sym_dev | 0.2504 | 2.875 | 0.1808 | 2.477 |
| Drop_asym_peer | 0.2437 | 2.860 | 0.1747 | 2.430 |
| Drop_asym_dev | 0.2604 | 2.938 | 0.1768 | 2.510 |

The table reports out-of-sample performance of drop-one-out PI indices based on the shared-analyst or customer–supplier networks. In each specification, we exclude one component (in the case of multiple characteristics, all transformed variables for $s = 1, \ldots, S$ within that category) and construct the PI index using the remaining three components. Out-of-sample $R^2$ (%) is computed using a rolling estimation window of 84 months, as described in Section 3.2. The long–short equal-weighted portfolios are formed following the procedure in Section 3.3.

- In asymmetric networks (shared analysts; customer–supplier), the antisymmetric peer-average component is especially valuable: dropping it hurts the most.
- Dropping the antisymmetric deviation can even *improve* performance, suggesting it may add noise in some settings.

## 10 Appendix tables (12–16): what they add

- **Table 12:** the complete list of characteristics and their categorizations. Key message: PI is built from a *broad* set of signals, not a narrow anomaly.
- **Table 13:** descriptive statistics of characteristics and their correlations with PI. Key message: correlations are generally modest; PI is not trivially one characteristic in disguise.
- **Table 14:** descriptive statistics for peer-average and deviation variables used in PI construction and average ridge coefficients. Key message: both peer averages and deviations receive non-trivial weights, consistent with the "two-channel" structure.
- **Table 15:** PI abnormal returns are not explained away by simple ETF exposures to Quality/Value/Momentum; alpha remains.
- **Table 16:** sentiment variables do not subsume PI; PI remains significant.

## 11 Consolidated takeaways

1. **Conceptual contribution:** expected returns can depend on *peer-transformed* characteristics, not only on own characteristics. This yields two interpretable channels: peer strength and relative position.
2. **Empirical contribution:** PI predicts returns and earnings surprises at short and long horizons, is robust to many controls, and is not subsumed by own-characteristic ML forecasts.
3. **Mechanism evidence:** dynamics and heterogeneity strongly point to *underreaction / slow information diffusion* in peer information.

**Table 12**
Characteristics and categorizations.

| Acronym | Characteristic name | Category |
|---|---|---|
| age | Firm age | Low_Leverage |
| aliq_at | Liquidity of book assets | Investment |
| aliq_mat | Liquidity of market assets | Low_Leverage |
| ami_126d | Amihud Measure | Size |
| at_be | Book leverage | Low_Leverage |
| at_gr1 | Asset Growth | Investment |
| at_me | Assets-to-market | Value |
| at_turnover | Capital turnover | Quality |
| be_gr1a | Change in common equity | Investment |
| be_me | Book-to-market equity | Value |
| beta_60m | Market Beta | Low_Risk |
| beta_dimson_21d | Dimson beta | Low_Risk |
| betadown_252d | Downside beta | Low_Risk |
| bev_mev | Book-to-market enterprise value | Value |
| bidaskhl_21d | High–low bid–ask spread | Low_Leverage |
| capx_gr1 | CAPEX growth (1 yr) | Investment |
| capx_gr2 | CAPEX growth (2 yrs) | Investment |
| capx_gr3 | CAPEX growth (3 yrs) | Investment |
| cash_at | Cash-to-assets | Low_Leverage |
| chcsho_12m | Net stock issues | Value |
| coa_gr1a | Change in current operating assets | Investment |
| col_gr1a | Change in current operating liabilities | Investment |
| coskew_21d | Coskewness (21 d) | Seasonality |
| cowc_gr1a | Change in current operating working capital | Accruals |
| cop_at | Cash-based operating profits-to-book assets | Quality |
| cop_atl1 | Cash-based operating profits-to-lagged book assets | Quality |
| dgp_dsale | Change gross margin minus change sales | Quality |
| dbnetis_at | Debt-net issues-to-assets | Seasonality |
| debt_gr3 | Growth in book debt (3 yrs) | Debt_Issuance |
| debt_me | Book debt-to-market equity | Value |
| div12m_me | Dividend yield | Value |
| dolvol_126d | Dollar trading volume (126 d) | Size |
| dolvol_var_126d | CV of dollar trading volume (126 d) | Profitability |
| dsale_drec | Change sales minus change receivables | Profit_Growth |
| dsale_dsga | Change sales minus change SG&A | Profit_Growth |
| ebit_bev | EBIT-to-book enterprise value | Profitability |
| ebit_sale | EBIT-to-sales | Profitability |
| ebitda_mev | EBITDA-to-market enterprise value | Value |
| emp_gr1 | Employment growth | Investment |
| eq_dur | Equity duration | Value |
| eqnetis_at | Equity-net issues-to-assets | Value |
| eqnpo_12m | Equity net payout (12 m) | Value |
| eqnpo_me | Equity net payout-to-market equity | Value |
| eqpo_me | Equity purchases-to-market equity | Value |
| fcf_me | Free cash flow-to-market equity | Value |
| fnl_gr1a | Change in financial liabilities | Debt_Issuance |
| f_score | Piotroski F-score | Profitability |
| gp_at | Gross profits-to-assets | Quality |
| gp_atl1 | Gross profits-to-lagged assets | Quality |
| inv_gr1 | Inventory growth | Investment |
| inv_gr1a | Change in inventory | Investment |
| iskew_capm_21d | Idiosyncratic skewness (CAPM, 21 d) | Short_Term_Reversal |
| iskew_ff3_21d | Idiosyncratic skewness (FF3, 21 d) | Short_Term_Reversal |
| iskew_hxz4_21d | Idiosyncratic skewness (HXZ4, 21 d) | Short_Term_Reversal |
| ivol_capm_21d | Idiosyncratic volatility (CAPM, 21 d) | Low_Risk |
| ivol_capm_252d | Idiosyncratic volatility (CAPM, 252 d) | Low_Risk |
| ivol_ff3_21d | Idiosyncratic volatility (FF3, 21 d) | Low_Risk |
| ivol_hxz4_21d | Idiosyncratic volatility (HXZ4, 21 d) | Low_Risk |
| kz_index | Kaplan–Zingales index | Seasonality |
| lnoa_gr1a | Change in long-term net operating assets | Investment |
| lti_gr1a | Change in long-term investments | Seasonality |
| market_equity | Market equity | Size |
| mispricing_mgmt | Change in net operating assets (mispricing) | Investment |
| mispricing_perf | Mispricing performance | Quality |
| ncoa_gr1a | Change in noncurrent operating assets | Investment |
| ncol_gr1a | Change in noncurrent operating liabilities | Debt_Issuance |
| netdebt_me | Net debt-to-market equity | Low_Leverage |
| netis_at | Net issues-to-assets | Value |
| nfna_gr1a | Change in net financial assets | Debt_Issuance |
| ni_be | Net income-to-book equity | Profitability |
| ni_inc8q | Change in quarterly return on income (8 q) | Quality |
| ni_me | Net income-to-market equity | Value |
| niq_at | Quarterly net income-to-assets | Quality |

(continued on next page)

**Table 12** (continued).

| | | |
|---|---|---|
| niq_at_chg1 | Change in quarterly ROA | Profit_Growth |
| niq_be | Quarterly net income-to-book equity | Profitability |
| niq_be_chg1 | Change in quarterly ROE | Profit_Growth |
| niq_su | Net income surprise | Profit_Growth |
| nncoa_gr1a | Change in net noncurrent operating assets | Investment |
| noa_at | Net operating assets-to-assets | Debt_Issuance |
| noa_gr1a | Change in net operating assets | Investment |
| ocf_at | Operating cash flow-to-assets | Profitability |
| ocf_at_chg1 | Change in operating cash flow-to-assets | Profit_Growth |
| ocf_me | Operating cash flow-to-market equity | Value |
| ocfq_saleq_std | Std. dev. of quarterly OCF-to-sales | Low_Risk |
| oaccruals_at | Operating accruals-to-assets | Accruals |
| oaccruals_ni | Operating accruals-to-net income | Accruals |
| o_score | Ohlson O-score | Profitability |
| op_at | Operating profits-to-assets | Quality |
| op_atl1 | Operating profits-to-lagged assets | Quality |
| ope_be | Cash-based operating profits-to-book equity | Profitability |
| ope_bel1 | Cash-based operating profits-to-lagged book equity | Profitability |
| opex_at | Operating expenses-to-assets | Quality |
| ppeinv_gr1a | Change PPE and Inventory | Investment |
| prc | Price per share | Size |
| prc_highprc_252d | 12-month maximum price ratio | Momentum |
| qmj_prof | QMJ profitability factor | Quality |
| qmj_safety | QMJ safety factor | Quality |
| resff3_6_1 | FF3 residual return (6–1 m) | Momentum |
| resff3_12_1 | FF3 residual return (12–1 m) | Momentum |
| ret_1_0 | 1-month reversal return | Short_Term_Reversal |
| ret_3_1 | 3–1-month momentum return | Momentum |
| ret_6_1 | 6–1-month momentum return | Momentum |
| ret_9_1 | 9–1-month momentum return | Momentum |
| ret_12_1 | 12–1-month momentum return | Momentum |
| ret_12_7 | 12–7-month return | Profit_Growth |
| rmax1_21d | 1-day maximum return (21 d) | Low_Risk |
| rmax5_21d | 5-day maximum return (21 d) | Low_Risk |
| rmax5_rvol_21d | Max-return-to-volatility ratio (21 d) | Short_Term_Reversal |
| rskew_21d | Return skewness (21 d) | Short_Term_Reversal |
| rvol_21d | Return volatility (21 d) | Low_Risk |
| sale_bev | Sales-to-book enterprise value | Quality |
| sale_emp_gr1 | Sales-per-employee growth | Profit_Growth |
| sale_gr1 | Sales growth (1 yr) | Investment |
| sale_gr3 | Sales growth (3 yrs) | Investment |
| sale_me | Sales-to-market equity | Value |
| saleq_gr1 | Sales growth (1 q) | Investment |
| saleq_su | Sales surprise | Profit_Growth |
| seas_1_1an | Annual seasonal return | Profit_Growth |
| seas_1_1na | NA seasonal return | Momentum |
| sti_gr1a | Change in short-term investments | Seasonality |
| taccruals_at | Total accruals-to-assets | Accruals |
| taccruals_ni | Total accruals-to-net income | Accruals |
| tangibility | Asset tangibility | Low_Leverage |
| tax_gr1a | Change in tax expenses | Profit_Growth |
| turnover_126d | Share turnover (126 d) | Low_Risk |
| turnover_var_126d | CV of turnover (126 d) | Profitability |
| z_score | Altman Z-score | Low_Leverage |
| zero_trades_21d | Zero-volume days (21 d) | Low_Risk |
| zero_trades_126d | Zero-volume days (126 d) | Low_Risk |
| zero_trades_252d | Zero-volume days (252 d) | Low_Risk |

This table lists all the 130 firm-specific characteristics we study in this paper. For the source of each characteristic, please see Jensen et al. (2023). The firm-specific characteristics are classified into 13 groups: Accruals, Debt_Issuance, Investment, Low_Leverage, Low_Risk, Momentum, Profit_Growth, Profitability, Quality, Seasonality, Short_Term_Reversal, Size, and Value.

**Table 13**
Descriptive statistics—raw values.

| Variable | Observations (in 000's) | Mean | Standard deviation | Correlation with PI |
|---|---|---|---|---|
| (1) Accruals | | | | |
| Change in current operating working capital(cowc_gr1a) | 2 394 768 | 0.004 | 0.186 | −0.066 |
| Operating accruals(oaccruals_at) | 2 568 257 | −0.053 | 0.377 | −0.045 |
| Percent operating accruals(oaccruals_ni) | 2 566 614 | −1.647 | 44.739 | −0.018 |
| Total accruals(taccruals_at) | 2 540 945 | −0.061 | 0.473 | 0.001 |
| Percent total accruals(taccruals_ni) | 2 538 887 | −3.225 | 81.831 | 0.006 |
| (2) Debt_Issuance | | | | |
| Growth in book debt (3 years)(debt_gr3) | 2 039 953 | 17.753 | 438.973 | −0.014 |
| Change in financial liabilities(fnl_gr1a) | 2 685 962 | 0.012 | 0.287 | −0.063 |
| Change in noncurrent operating liabilities(ncol_gr1a) | 2 420 052 | 0.003 | 0.105 | −0.004 |
| Change in net financial assets(nfna_gr1a) | 2 685 962 | −0.009 | 0.304 | 0.052 |
| Net operating assets(noa_at) | 2 396 636 | 1.311 | 33.504 | −0.009 |
| (3) Investment | | | | |
| Liquidity of book assets(aliq_at) | 2 423 731 | 1.518 | 29.948 | −0.010 |
| Asset Growth(at_gr1) | 2 691 394 | 1.246 | 44.363 | −0.010 |
| Change in common equity(be_gr1a) | 2 538 328 | 0.020 | 0.259 | −0.085 |
| CAPEX growth (1 year)(capx_gr1) | 2 425 229 | 1.756 | 38.721 | −0.018 |
| CAPEX growth (2 years)(capx_gr2) | 2 247 680 | 4.013 | 101.770 | −0.017 |
| CAPEX growth (3 years)(capx_gr3) | 2 071 885 | 5.322 | 122.147 | −0.017 |
| Change in current operating assets(coa_gr1a) | 2 418 001 | 0.019 | 0.188 | −0.084 |
| Change in current operating liabilities(col_gr1a) | 2 428 113 | 0.014 | 0.151 | −0.028 |
| Hiring rate(emp_gr1) | 2 279 786 | 0.062 | 0.313 | −0.118 |
| Inventory growth(inv_gr1) | 2 035 779 | 0.665 | 13.097 | −0.019 |
| Inventory change(inv_gr1a) | 2 641 962 | 0.006 | 0.080 | −0.093 |
| Change in long-term net operating assets(lnoa_gr1a) | 2 312 163 | 0.028 | 0.120 | −0.142 |
| Mispricing factor: Management(mispricing_mgmt) | 2 680 207 | 0.499 | 0.183 | 0.246 |
| Change in noncurrent operating assets(ncoa_gr1a) | 2 418 991 | 0.029 | 0.298 | −0.098 |
| Change in net noncurrent operating assets(nncoa_gr1a) | 2 387 718 | 0.025 | 0.306 | −0.095 |
| Change in net operating assets(noa_gr1a) | 2 386 445 | 0.809 | 33.785 | −0.009 |
| Change PPE and Inventory(ppeinv_gr1a) | 2 363 836 | 0.157 | 3.325 | −0.024 |
| Sales Growth (1 year)(sale_gr1) | 2 587 505 | 0.653 | 15.245 | −0.019 |
| Sales Growth (3 years)(sale_gr3) | 2 228 502 | 3.630 | 105.998 | −0.010 |
| Sales growth (1 quarter)(saleq_gr1) | 2 558 116 | 0.819 | 19.837 | −0.013 |
| (4) Low_Leverage | | | | |
| Firm age(age) | 2 980 487 | 212.785 | 190.960 | 0.057 |
| Liquidity of market assets(aliq_mat) | 2 252 927 | 0.494 | 0.330 | 0.136 |
| Book leverage(at_be) | 2 708 629 | 5.111 | 38.977 | −0.021 |
| The high-low bid–ask spread(bidaskhl_21d) | 2 671 495 | 0.020 | 0.036 | −0.009 |
| Cash-to-assets(cash_at) | 2 797 093 | 0.180 | 0.227 | 0.016 |
| Net debt-to-price(netdebt_me) | 2 795 312 | 0.676 | 4.888 | −0.014 |
| Asset tangibility(tangibility) | 2 446 681 | 0.649 | 0.235 | 0.076 |
| Altman Z-score(z_score) | 2 406 981 | 6.398 | 42.079 | −0.037 |
| (5) Low_Risk | | | | |
| Market Beta(beta_60m) | 2 220 915 | 1.122 | 0.758 | −0.009 |
| Dimson beta(beta_dimson_21d) | 2 616 482 | 0.889 | 2.507 | −0.024 |
| Downside beta(betadown_252d) | 2 546 391 | 0.924 | 0.894 | −0.059 |
| Idiosyncratic volatility from the CAPM (21 days)(ivol_capm_21d) | 2 616 482 | 0.033 | 0.031 | −0.111 |
| Idiosyncratic volatility from the CAPM (252 days)(ivol_capm_252d) | 2 578 926 | 0.036 | 0.027 | −0.086 |
| Idiosyncratic volatility from the Fama–French 3-factor model(ivol_ff3_21d) | 2 616 482 | 0.032 | 0.031 | −0.106 |
| Idiosyncratic volatility from the q-factor model(ivol_hxz4_21d) | 2 616 482 | 0.032 | 0.031 | −0.106 |
| Cash flow volatility(ocfq_saleq_std) | 2 021 187 | 5.339 | 81.857 | −0.026 |
| Maximum daily return(rmax1_21d) | 2 616 482 | 0.082 | 0.104 | −0.136 |
| Highest 5 days of return(rmax5_21d) | 2 616 482 | 0.046 | 0.044 | −0.167 |
| Return volatility(rvol_21d) | 2 616 482 | 0.035 | 0.032 | −0.115 |
| Share turnover(turnover_126d) | 2 820 802 | 0.006 | 0.026 | −0.050 |
| Number of zero trades ...(1 month)(zero_trades_21d) | 2 870 711 | 1.484 | 3.657 | 0.054 |
| Number of zero trades ...(6 months)(zero_trades_126d) | 2 820 802 | 1.467 | 3.436 | 0.070 |
| Number of zero trades ...(12 months)(zero_trades_252d) | 2 753 339 | 1.449 | 3.350 | 0.073 |
| (6) Momentum | | | | |
| Current price to high price over last year(prc_highprc_252d) | 2 577 361 | 0.734 | 0.231 | 0.093 |
| Residual momentum t-6 to t-1(resff3_6_1) | 2 422 411 | −0.072 | 0.648 | 0.084 |
| Residual momentum t-12 to t-1(resff3_12_1) | 2 425 026 | −0.029 | 0.285 | 0.231 |
| Price momentum t-3 to t-1(ret_3_1) | 2 930 213 | 0.021 | 0.267 | 0.017 |
| Price momentum t-6 to t-1(ret_6_1) | 2 867 342 | 0.053 | 0.450 | 0.058 |
| Price momentum t-9 to t-1(ret_9_1) | 2 804 836 | 0.088 | 0.599 | 0.098 |
| Price momentum t-12 to t-1(ret_12_1) | 2 742 959 | 0.128 | 0.747 | 0.112 |
| Year 1-lagged return, non-annual(seas_1_1na) | 2 704 009 | 0.011 | 0.055 | 0.023 |

**Table 13** (continued).

| | | | | |
|---|---|---|---|---|
| (7) Profit_Growth | | | | |
| Change sales minus change receivables(dsale_drec) | 2 347 352 | −0.180 | 14.578 | 0.006 |
| Change sales minus change SG&A(dsale_dsga) | 2 013 334 | 0.125 | 3.832 | −0.014 |
| Change in quarterly return on assets(niq_at_chg1) | 2 461 044 | 0.007 | 0.470 | 0.004 |
| Change in quarterly return on equity(niq_be_chg1) | 2 331 291 | −0.023 | 2.049 | 0.011 |
| Standardized earnings surprise(niq_su) | 2 256 233 | −0.183 | 3.096 | 0.096 |
| Change in operating cash flow to assets(ocf_at_chg1) | 2 607 088 | 0.008 | 0.454 | 0.015 |
| Price momentum t-12 to t-7(ret_12_7) | 2 736 461 | 0.057 | 0.443 | 0.097 |
| Labor force efficiency(sale_emp_gr1) | 2 341 645 | 0.396 | 8.629 | −0.014 |
| Standardized revenue surprise(saleq_su) | 2 212 543 | 0.164 | 2.838 | 0.044 |
| Year 1-lagged return, annual(seas_1_1an) | 2 713 456 | 0.012 | 0.178 | 0.089 |
| Tax expense surprise(tax_gr1a) | 2 653 815 | 0.002 | 0.042 | 0.031 |
| (8) Profitability | | | | |
| Coefficient of variation for dollar trading volume(dolvol_var_126d) | 2 820 732 | 1.458 | 1.042 | 0.024 |
| Return on net operating assets(ebit_bev) | 2 659 302 | −0.700 | 17.905 | 0.013 |
| Profit margin(ebit_sale) | 2 719 724 | −4.239 | 102.090 | 0.019 |
| Piotroski F-score(f_score) | 2 149 003 | 4.778 | 1.788 | 0.147 |
| Return on equity(ni_be) | 2 696 790 | −0.356 | 8.276 | 0.029 |
| Quarterly return on equity(niq_be) | 2 559 057 | −0.049 | 1.629 | 0.035 |
| Ohlson O-score(o_score) | 2 357 260 | −1.908 | 6.094 | −0.074 |
| Operating cash flow to assets(ocf_at) | 2 744 827 | −0.020 | 0.357 | 0.125 |
| Operating profits-to-book equity(ope_be) | 2 301 662 | −0.066 | 6.225 | 0.033 |
| Operating profits-to-lagged book equity(ope_bel1) | 2 192 465 | 0.042 | 7.541 | 0.024 |
| Coefficient of variation for share turnover(turnover_var_126d) | 2 820 738 | 1.427 | 1.017 | 0.033 |
| (9) Quality | | | | |
| Capital turnover(at_turnover) | 2 760 086 | 0.991 | 0.945 | 0.049 |
| Cash-based operating profits-to-book assets(cop_at) | 2 566 079 | 0.124 | 0.365 | 0.144 |
| Cash-based operating profits-to-lagged book assets(cop_atl1) | 2 537 857 | 0.003 | 9.764 | 0.010 |
| Change gross margin minus change sales(dgp_dsale) | 2 282 887 | 0.005 | 8.787 | 0.014 |
| Gross profits-to-assets(gp_at) | 2 785 344 | 0.276 | 0.360 | 0.109 |
| Gross profits-to-lagged assets(gp_atl1) | 2 682 928 | 0.339 | 0.970 | 0.020 |
| Mispricing factor: Performance(mispricing_perf) | 2 906 714 | 0.496 | 0.213 | 0.217 |
| Number of consecutive quarters with earnings increases(ni_inc8q) | 2 103 334 | 2.735 | 3.112 | 0.031 |
| Quarterly return on assets(niq_at) | 2 660 465 | −0.016 | 0.197 | 0.082 |
| Operating profits-to-book assets(op_at) | 2 786 120 | 0.067 | 0.292 | 0.122 |
| Operating profits-to-lagged book assets(op_atl1) | 2 683 679 | 0.056 | 1.618 | 0.039 |
| Operating leverage(opex_at) | 2 785 477 | 0.937 | 0.943 | 0.036 |
| Quality minus Junk: Profitability(qmj_prof) | 2 775 068 | 0.002 | 1.000 | 0.183 |
| Quality minus Junk: Safety(qmj_safety) | 2 814 590 | 0.000 | 0.997 | 0.094 |
| Assets turnover(sale_bev) | 2 659 462 | 3.111 | 25.085 | 0.016 |
| (10) Seasonality | | | | |
| Coskewness(coskew_21d) | 2 616 482 | −0.014 | 0.315 | −0.011 |
| Net debt issuance(dbnetis_at) | 2 772 222 | −0.037 | 0.299 | −0.022 |
| Kaplan–Zingales index(kz_index) | 2 424 265 | −30.214 | 597.785 | 0.007 |
| Change in long-term investments(lti_gr1a) | 2 488 809 | 0.004 | 0.077 | −0.022 |
| Change in short-term investments(sti_gr1a) | 2 448 288 | 0.003 | 0.128 | −0.030 |
| (11) Short_Term_Reversal | | | | |
| Idiosyncratic skewness from the CAPM(iskew_capm_21d) | 2 616 482 | 0.242 | 0.947 | −0.128 |
| Idiosyncratic skewness from the Fama–French 3-factor model(iskew_ff3_21d) | 2 616 482 | 0.201 | 0.860 | −0.103 |
| Idiosyncratic skewness from the q-factor model(iskew_hxz4_21d) | 2 616 482 | 0.182 | 0.814 | −0.098 |
| Short-term reversal(ret_1_0) | 2 953 743 | 0.010 | 0.190 | −0.235 |
| Highest 5 days of return scaled by volatility(rmax5_rvol_21d) | 2 479 523 | 1.218 | 0.539 | −0.210 |
| Total skewness(rskew_21d) | 2 616 482 | 0.242 | 0.951 | −0.143 |
| (12) Size | | | | |
| Amihud Measure(ami_126d) | 2 605 890 | 7.837 | 66.724 | 0.005 |
| Dollar trading volume(dolvol_126d) | 2 820 796 | 19 410 405.633 | 134 964 837.005 | −0.011 |
| Market equity(characteristic)(market_equity) | 2 976 197 | 2 759.602 | 18 394.302 | 0.003 |
| Price per share in dollars(prc) | 2 976 197 | 30.178 | 707.910 | −0.002 |
| (13) Value | | | | |
| Assets-to-market(at_me) | 2 800 937 | 3.484 | 14.484 | 0.016 |
| Book-to-market equity(be_me) | 2 706 306 | 0.836 | 1.457 | 0.087 |
| Book-to-market enterprise value(bev_mev) | 2 638 084 | 0.814 | 3.096 | 0.059 |
| Net stock issues(chcsho_12m) | 2 728 645 | 0.160 | 2.458 | −0.032 |
| Debt-to-market(debt_me) | 2 795 312 | 1.004 | 5.470 | 0.000 |
| Dividend yield(div12m_me) | 2 763 825 | 0.013 | 0.052 | 0.041 |
| Ebitda-to-market enterprise value(ebitda_mev) | 2 725 915 | 0.009 | 2.117 | −0.003 |
| Equity duration(eq_dur) | 2 331 957 | 17.696 | 37.033 | −0.015 |
| Net equity issuance(eqnetis_at) | 2 641 812 | 0.075 | 0.271 | −0.152 |
| Equity net payout(eqnpo_12m) | 2 712 461 | −0.048 | 0.260 | 0.128 |
| Net payout yield(eqnpo_me) | 2 634 247 | −0.043 | 0.411 | 0.041 |
| Payout yield(eqpo_me) | 2 483 779 | 0.035 | 0.156 | 0.031 |

**Table 13** (continued).

| | | | | |
|---|---|---|---|---|
| Free cash flow-to-price(fcf_me) | 2 616 751 | −0.112 | 1.067 | 0.035 |
| Net total issuance(netis_at) | 2 641 485 | 0.035 | 0.432 | −0.110 |
| Earnings-to-price(ni_me) | 2 789 808 | −0.149 | 1.618 | −0.011 |
| Operating cash flow-to-market(ocf_me) | 2 742 194 | 0.011 | 0.906 | 0.042 |
| Sales-to-market(sale_me) | 2 783 446 | 2.147 | 6.440 | 0.051 |
| Index | | | | |
| Peer Index(PI) | 2,557,030 | 0.000 | 0.289 | 1.000 |

The table displays descriptive statistics for firm characteristics defined in Table 12 and the new PI index. The sample spans the period January 1980 through Dec 2024. Means, standard deviations, and correlations are based on the original (unstandardized) variables. To mitigate the influence of outliers, we winsorize the data at the 0.01% and 99.99% levels when computing means and standard deviations. However, our main empirical analysis — following the data filtering procedures of Didisheim et al. (2024) and Kelly et al. (2025) — does not apply winsorization.

**Table 14** (continued).

| | | | |
|---|---|---|---|
| eqpo_me_peer | 0.011 | inv_gr1_dev | 0.016 |
| dsale_dsga_peer | 0.01 | ope_be_dev | 0.016 |
| tangibility_peer | 0.009 | iskew_capm_21d_dev | 0.015 |
| coa_gr1a_peer | 0.009 | kz_index_dev | −0.015 |
| ivol_hxz4_21d_peer | 0.008 | rskew_21d_dev | −0.015 |
| eqnpo_me_peer | −0.008 | betadown_252d_dev | 0.014 |
| ncol_gr1a_peer | −0.008 | netis_at_dev | −0.013 |
| kz_index_peer | −0.008 | ni_be_dev | 0.013 |
| turnover_126d_peer | 0.008 | f_score_dev | −0.013 |
| iskew_ff3_21d_peer | −0.006 | ni_inc8q_dev | −0.01 |
| lnoa_gr1a_peer | −0.006 | cop_at_dev | 0.006 |
| netdebt_me_peer | −0.005 | niq_at_dev | 0.005 |
| fnl_gr1a_peer | 0.004 | capx_gr3_dev | 0.005 |
| ppeinv_gr1a_peer | −0.003 | nfna_gr1a_dev | −0.004 |
| lti_gr1a_peer | −0.002 | ocf_at_chg1_dev | −0.002 |
| rskew_21d_peer | −0.002 | lnoa_gr1a_dev | −0.002 |
| oaccruals_ni_peer | −0.002 | eqnpo_me_dev | −0.002 |

The variables are presented in descending order by the magnitude of average coefficients in all the estimation windows. The column "Avg Coef" presents the average coefficient*100 for each variable.

**Table 15**
Do relevant ETFs explain PI abnormal returns?.

| | Excess ret | Alpha | Quality | Value | Momentum |
|---|---|---|---|---|---|
| Top-minus-bottom portfolio | 0.92*** | | | | |
| | (6.13) | | | | |
| | | 0.63** | −0.24*** | | |
| | | (2.21) | (−3.75) | | |
| | | 0.51** | | −0.14*** | |
| | | (2.49) | | (−3.27) | |
| | | 0.60** | | | −0.20*** |
| | | (2.09) | | | (−3.22) |
| | | 0.70** | −0.82*** | 0.56*** | 0.08 |
| | | (2.51) | (−3.92) | (3.51) | (0.65) |

This table reports the loading coefficients and alphas obtained from regressing monthly value-weighted top-minus-bottom PI portfolio returns on BlackRock's iShares MSCI USA Quality Factor ETF (Quality), iShares S&P 500 Value ETF (Value), and iShares MSCI USA Momentum Factor ETF (Momentum). Monthly returns of Quality, Value, and Momentum are available from August 2013, June 2000, and June 2013, respectively. The sample period ends in Dec 2024. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

**Table 16**
Sentiment effect and PI.

| Dependent variable | Observations | Stock sentiment | Industry sentiment | PI |
|---|---|---|---|---|
| $R_{t+1}$ | 1,032,059 | 0.66 | | |
| | | (1.12) | | |
| $R_{t+1}$ | 1,032,059 | | −0.21 | |
| | | | (−0.57) | |
| $R_{t+1}$ | 1,032,059 | 0.71 | −0.42 | 0.53*** |
| | | (1.11) | (−1.06) | (3.10) |

This table shows the average slopes*100; *t*-statistics with Newey–West adjustments of 12 lags from a set of monthly value-weighted Fama–MacBeth regressions. The stock sentiment and industry sentiment variables represent the stock-level sentiment index and industry-level sentiment index, respectively, constructed using RavenPack News Analytics. Control variables include size and book-to-market ratio. Our sample period starts from January 2000 when the news dataset became available. One, two, and three asterisks indicate significance at the 10%, 5%, and 1% levels, respectively.

4. **Network perspective:** directed networks naturally create a symmetric–antisymmetric decomposition, and the antisymmetric (directional) peer-average component can be a distinct and powerful source of predictability.

# 12 Conclusion

## 12.1 Summary

The paper's central conclusion is that a large and economically meaningful component of cross-sectional return predictability is *network-structured*: information from *economically linked peers* helps forecast a firm's future returns and fundamentals, and this information is not fully captured by models that rely only on a firm's own characteristics.

The authors operationalize this insight by constructing a **Peer Index (PI)** that maps peer information into a stock's expected return through two interpretable channels:

$$z_{i,t} \equiv \text{(peer-average characteristics: “peer strength”)}, \tag{2}$$

$$\tilde{x}_{i,t} \equiv x_{i,t} - z_{i,t} \quad \text{(deviation from peers: “relative position”)}, \tag{3}$$

$$PI_{i,t} \equiv z'_{i,t}\widehat{\beta}_z + \tilde{x}'_{i,t}\widehat{\beta}_x \qquad \text{(estimated via ridge).} \tag{4}$$

This dual-channel construction matters because it clarifies *why* one firm's characteristics can predict another firm's return: firms inherit information from their peer groups, and being unusually positioned within a peer group is informative on top of the group mean.

## 12.2 Key empirical conclusions

1. **PI predicts returns strongly and robustly.** Portfolio sorts on PI generate large and statistically strong spreads. For example (Table 3), the monthly top-minus-bottom PI portfolio earns about

   $$\text{EW: } 3.33\% \text{ (t=23.03)}, \qquad \text{VW: } 0.89\% \text{ (t=6.05)},$$

   with large factor-model alphas (seven-factor model):

   $$\alpha_{\text{EW}} \approx 3.18\% \text{ (t=23.60)}, \qquad \alpha_{\text{VW}} \approx 0.68\% \text{ (t=5.45)}.$$

   Even after removing the component of PI that is predictable by an own-characteristics ML forecast, the remaining “ML-adjusted PI” still delivers strong spreads (Table 3).

2. **PI adds information beyond powerful ML predictors built from firm-level characteristics.** Out-of-sample (Table 2), PI alone achieves $R^2_{\text{OOS}} \approx 0.325\%$ and combining PI with ML forecasts improves forecasting accuracy further:

   $$R^2_{\text{OOS}}(\text{ML+PI}) \in [0.391\%, 0.498\%]$$

   depending on the ML benchmark. This supports the interpretation that PI captures *genuine cross-stock information* that does not reduce to a better nonlinear transformation of $x_{i,t}$ alone.

3. **PI forecasts fundamentals (earnings surprises), not only returns.** PI predicts industry-level earnings surprises and industry returns (Table 6) and also predicts firm-level earnings surprises at one-, four-, and eight-quarter horizons (Table 7), with PI remaining positive at longer horizons even when ML signals weaken or flip sign. This links the return predictability to slow-moving information about cash flows.

4. **The dynamics support underreaction / slow diffusion rather than reversal.** Lag-augmented local projections (Fig. 3) show that positive PI innovations are followed by higher future returns that *gradually decay* toward zero *without a reversal*. This time-series pattern is consistent with sluggish incorporation of peer information into prices.

5. **The effect is stronger in weak information environments (mechanism evidence).** Heterogeneity tests (Table 8) show larger PI effects when information processing is plausibly harder or attention is weaker: smaller firms, fewer analysts, higher analyst dispersion, fewer news mentions, and smaller changes in news coverage.

6. **The effect is not easily explained by standard risk compensation or sentiment.** The paper presents spanning and robustness evidence indicating that PI's predictive content is not simply a repackaging of standard factor exposures, and sentiment-based controls do not eliminate the PI effect.

### 12.3 Intuition: why PI works

A coherent economic interpretation is **limited attention to peer-relevant information**. Investors (and, to some extent, analysts) overweight firm-specific signals and underweight information embedded in economically linked peers. Because peer information is diffuse (it requires aggregation across linked firms and interpretation of relative positioning), it enters prices gradually.

The PI construction is therefore powerful because it explicitly encodes the two pieces of information that are naturally slow to process:

- **Peer strength (average channel):** if the peer group exhibits strong profitability, investment opportunities, or other value-relevant characteristics, a focal firm is partly exposed to that state through shared economic conditions and linkages.

- **Relative position (deviation channel):** if a firm is unusually cheap, unusually constrained, or unusually profitable relative to peers, this contains incremental information (mispricing or future fundamentals) beyond the peer-group average.

Because markets do not instantly process these network-based signals, PI predicts both *future earnings surprises* and *future returns*, and the absence of long-horizon reversal aligns with underreaction rather than transient overreaction.

### 12.4 Directed networks and "alpha" vs "beta" (how to interpret the asymmetric-network evidence)

Using symmetric industry peers, the peer linkage matrix is symmetric, so the directional component is mechanically absent. When the peer network is genuinely directed (customer–supplier links or shared-analyst co-coverage), the symmetric–antisymmetric decomposition clarifies that part of PI can be interpreted as:

- a **factor-exposure ("beta")** component tied to symmetric peer averages, and
- a **factor-neutral directional ("alpha")** component tied to the antisymmetric part of the network.

Drop-one-out results (Table 11) indicate that, in asymmetric networks, the directional peer-average component can be especially important, highlighting that economically directed links can generate predictability beyond symmetric industry clustering.

### 12.5 Takeaway

*Cross-stock predictability is network-based: an economically structured peer index that combines peer strength and within-peer relative position forecasts returns and earnings, improves out-of-sample performance beyond own-stock ML predictors, and exhibits slow-decaying (non-reversing) dynamics consistent with underreaction to peer information.*